

RMO(INDIA) – 1992

Max Mark – 100

Time – 3 Hrs

Instructions:

- Calculators (in any form) and protractors are not allowed.
 - Ruler and compass allowed.
 - Answer all the questions.
1. Determine the set of integers n , for which $n^2 + 19n + 92$ is a square.
 2. If $\frac{1}{a} + \frac{1}{b} = \frac{1}{c}$, where a, b, c are positive integers with no common factor. Prove that $a + b$ is a square.
 3. Determine the largest 3 digit prime factor of integer ${}^{2000}C_{1000}$.
 4. ABCD is a cyclic quadrilateral with $AC \perp BD$; AC meets BD at E. Prove that $EA^2 + EB^2 + EC^2 + ED^2 = 4R^2$, where R is the radius of circumscribing circle.
 5. ABCD is a cyclic quadrilateral; x, y, z are the distances of A from the lines BD, BC and CD respectively. Prove that $\frac{BD}{x} = \frac{BC}{y} + \frac{CD}{z}$.
 6. ABCD is a quadrilateral and P, Q are midpoints of CD and AB. AP, DQ meets at X, and BP, CQ meets at Y. Prove that area $\triangle ADX$ + area $\triangle BCY$ = area quadrilateral PXQY.
 7. Prove that $1 < \frac{1}{1001} + \frac{1}{1002} + \frac{1}{1003} + \dots + \frac{1}{3001} < 1\frac{1}{3}$.
 8. Solve the system of equation in terms of parameter A.
 $(x + y)(x + y + z) = 18$, $(y + z)(x + y + z) = 30$, $(z + x)(x + y + z) = 2A$.
 9. The cyclic octagon ABCDEFGH has sides a, a, a, a, b, b, b, b respectively. Find the radius of the circle that circumscribe ABCDEFGH.

**